

Capacitor Charging and a Power On Reset for an Active Low Microcontroller

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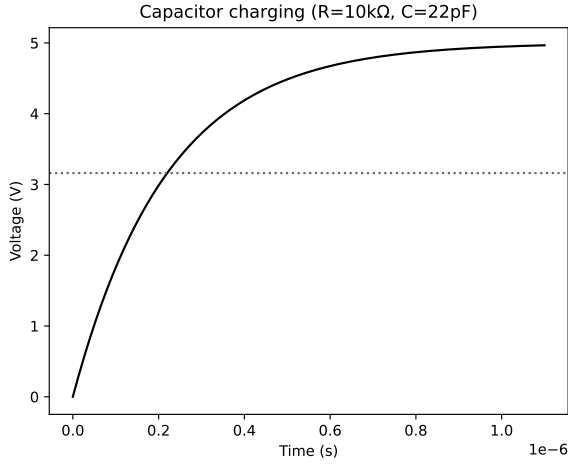


Fig. 1. Charging curves for four resistances, $C = 22$ pF.

Abstract—This document derives the charging behaviour of a resistor-capacitor (RC) network, the voltage reached after one time constant, and applies the result to size a power-on-reset (POR) circuit for a microcontroller with an active-low reset.

I. INTRODUCTION

When a capacitor C is charged through a series resistor R from a constant supply V_0 , the capacitor voltage rises according to

$$V(t) = V_0 \left(1 - e^{-t/RC}\right), \quad (1)$$

where $\tau = RC$ is the time constant. Substituting $t = \tau$ gives $V(\tau) = V_0(1 - e^{-1}) \approx 0.632 V_0$.

II. POWER ON RESET WITH AN ACTIVE LOW RESET

A power-on-reset circuit exploits (1) directly: R from V_{CC} to the reset node and C from that node to ground hold the microcontroller's active-low reset pin low until the node crosses V_{IH} , releasing the device.

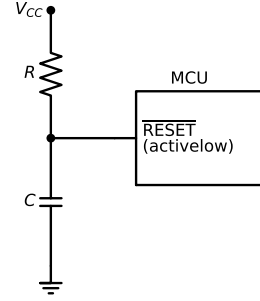


Fig. 2. Power-on-reset network driving an active-low reset pin.

A. Sizing R for a Timing Constraint

The reset pin releases the instant the node voltage reaches V_{IH} . Setting $V(t_{\text{release}}) = V_{IH}$ in (1) and solving for t_{release} :

$$\begin{aligned} V_{IH} &= V_0 \left(1 - e^{-t_{\text{release}}/RC}\right) \\ \frac{V_{IH}}{V_0} &= 1 - e^{-t_{\text{release}}/RC} \\ e^{-t_{\text{release}}/RC} &= 1 - \frac{V_{IH}}{V_0} \\ -\frac{t_{\text{release}}}{RC} &= \ln \left(1 - \frac{V_{IH}}{V_0}\right) \\ t_{\text{release}} &= -RC \ln \left(1 - \frac{V_{IH}}{V_0}\right). \end{aligned} \quad (2)$$

Rearranging (2) for R , given a target t_{release} and a chosen C :

$$R = \frac{-t_{\text{release}}}{C \ln \left(1 - \frac{V_{IH}}{V_0}\right)}. \quad (3)$$

1) *Worked example:* Take $V_{IH} = 0.7 V_0$, $V_0 = 5.0$ V, and a target $t_{\text{release}} = 10$ ms. The $C = 22$ pF used for the plots above was chosen only to keep the curves readable at a sub-microsecond timescale; a practical timing design instead starts from $C = 100$ nF. Evaluating the denominator of (3):

$$\ln(1 - 0.7) = \ln(0.3) \approx -1.204. \quad (4)$$

Substituting into (3):

$$R = \frac{-(10 \times 10^{-3})}{(100 \times 10^{-9}) \times (-1.204)} \approx 83 \text{ k}\Omega. \quad (5)$$

An 83 k Ω resistor with a 100 nF capacitor gives a reset hold-off of approximately 10 ms. Keeping C at the plotting value of

III. APPLICATIONS AND LIMITATIONS

A. Application

The RC-only power-on-reset circuit of Figure 2 is a reasonable design choice for a low-cost appliance controller board — for example, the control board inside a space heater, coffee maker, or garage-door opener — built around an 8-bit microcontroller running from a 5 V linear regulator.

An RC-only POR is acceptable here for three reasons. First, the regulated 5 V rail supplied by a linear regulator rises in well under a millisecond, so the step-input assumption behind the charging model of Section III holds; there is no slow, multi-second V_{CC} ramp to account for. Second, the appliance is either fully powered or fully off in normal use — there is no requirement to re-trigger a reset mid-operation, nor to catch a momentary dip in wall power, since a sag severe enough to matter would stop the appliance from functioning at all regardless of reset behaviour. Third, at production volumes of thousands of units, a two-component RC network is materially cheaper than a dedicated supervisor IC, and the appliance does not need that IC's precise threshold or brown-out detection to operate safely.

For this application, $R = 100\text{ k}\Omega$ (a 1% metal-film resistor) and $C = 1\text{ }\mu\text{F}$ (a $\pm 10\%$ X7R ceramic capacitor) were chosen. Using (6) with $V_0 = 5.0\text{ V}$ and $V_{IH} = 0.7 V_0$,

$$t_{\text{release}} = kRC = 1.204 \times (100,000)(1 \times 10^{-6}) = 120.4\text{ ms.} \quad (6)$$

This gives the microcontroller a comfortable 120 ms delay after power-up before it begins executing its initialization code.

B. Limitations

The RC-only approach has several limitations beyond component tolerance, addressed in Section III-C below.

- **No brown-out detection.** The circuit observes only the moment power first rises above V_{IH} . A later sag on the 5 V rail — for instance, an inductive load such as a relay coil switching on and briefly loading the supply — is invisible to this circuit, and the microcontroller could brown out mid-operation with no protection.
- **Sensitivity to a slowly-rising supply.** The derivation of t_{release} assumes V_{CC} behaves as a step input. If this board were ever powered from a slower-rising supply — a large bulk input capacitor, or a soft-start regulator — the RC node's voltage would rise in step with V_{CC} rather than lagging behind a fixed step, and the actual release point would no longer match the value predicted by equation (6).
- **No software- or watchdog-triggered reset.** The RC network can only reset the microcontroller once, at power-on. Recovering from a firmware hang during normal operation requires either a full power cycle or additional watchdog hardware, which this circuit alone cannot provide.

C. Component Tolerance

Applying (7) to the chosen $R = 100\text{ k}\Omega$ ($\pm 1\%$) and $C = 1\text{ }\mu\text{F}$ ($\pm 10\%$),

$$\frac{\Delta t}{t} \approx \frac{\Delta R}{R} + \frac{\Delta C}{C} = 0.01 + 0.10 = 0.11, \quad (7)$$

so the true release time could fall anywhere in the range

$$t_{\text{release}} = 120.4\text{ ms} \times (1 \pm 0.11) \Rightarrow 107.2\text{ ms to } 133.6\text{ ms.} \quad (8)$$

Computing the fastest and slowest corners directly — taking R and C simultaneously at their minimum and maximum values, respectively — gives 107.3 ms and 133.8 ms, matching the first-order estimate above to within the second-order term dropped in (7). Figure 3 shows the three corresponding charging curves.

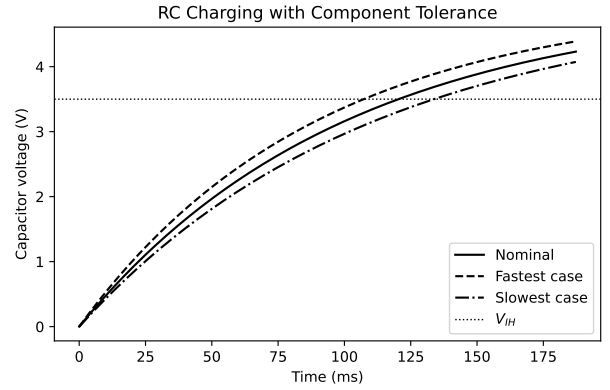


Fig. 3. Capacitor charging curves for the nominal, fastest-case, and slowest-case component values, with the V_{IH} threshold marked.